



TB HEALTHCARE SERVICE PROVIDER

To modernize TB laboratory operations in Tajikistan and Uzbekistan, Agaram Technologies implemented a distributed digital solution combining **Qualis LIMS, Qualis Portal, and Interfacer Middleware.**

This system replaced cumbersome manual workflows with automated, locally hosted LIMS instances that operated independently of internet connectivity, enabling uninterrupted sample testing and real-time data synchronization with central lab.

Integrated with diagnostic instruments, the system ensured accurate data capture, compliance with national regulations, and centralized reporting to health authorities, thus improving public healthcare in Central Asia.



WHAT'S UP WITH TUBERCULOSIS (TB) LABORATORIES

Laboratories run by a TB Healthcare Service Provider in Tajikistan and Uzbekistan traditionally relied on fragmented and manual systems to facilitate their operations.

Their challenges included,

- **Limited Digital Infrastructure:** Laboratories were using outdated systems or manual processes to track and report TB cases.
- **Lack of Data Centralization:** Data was being collected and stored in isolated systems, making it difficult to track the spread of TB at a national level.
- **Poor Internet Connectivity:** The internet infrastructure across regional laboratories was unreliable, leading to communication delays between central and regional laboratories.
- **Manual Data Exchange:** There was no efficient mechanism for exchanging critical TB data between labs, leading to delays in diagnosis, treatment, and reporting

HOW LAB SOFTWARE SOLUTIONS ACCELERATES HEALTHCARE

Q: What if you had a system to manage and track all TB lab operations from the first sample to final report?

A: Qualis LIMS streamlines laboratory workflows and ensures accurate, timely TB data collection, analysis, and reporting, customized to comply with national health regulations.

Q: What if you could automatically extract data from lab instruments?

A: Interfacer Middleware integrates instruments like GeneXpert, BACTEC MGIT, and BD-Epicenter directly to the LIMS, enabling seamless, real-time data capture and reducing errors.

Q: What if you had a secure, web-based way for labs to access, submit, and manage TB test orders and results?

A: The web-based portal, Qualis Portal, enables easy access to LIMS data for laboratory users, allowing real-time order submissions and results reporting.

GET THE PICTURE

With multiple TB laboratories spread across Tajikistan and Uzbekistan, a distributed system was essential to ensure seamless data exchange between regional and central labs despite unreliable internet connectivity. Laboratories faced challenges in collecting and sharing

TB-related data in a way that supported real-time reporting to national health authorities. Additionally, the diagnostic instruments used across the labs needed to be integrated into LIMS for effective data capture and analysis. To address these issues, a comprehensive system was required that could provide real-time access to data even in areas with limited or no internet availability.

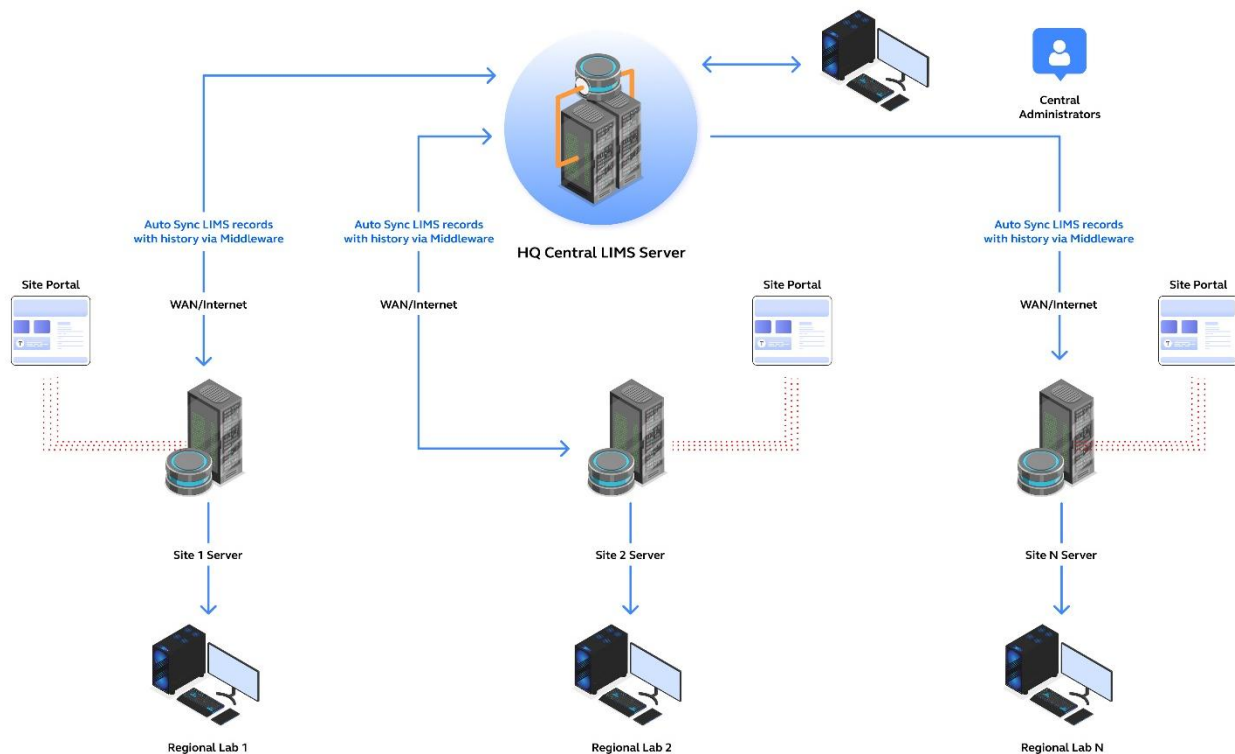
AUTOMATED TUBERCULOSIS (TB) LABORATORIES

To digitize and automate the management of TB laboratory workflows across the national public health, national reference, and regional culture laboratories in Tajikistan and Uzbekistan, we proposed the solution of a powerful combination of **Qualis LIMS**, **Qualis Portal** and **Interfacer Middleware**.

Agaram Technologies, in collaboration with a TB healthcare service provider and associates, began by conducting a detailed analysis of the TB laboratory systems. The team gathered information on the existing workflows, instruments in use, and reporting requirements to customize the solution effectively.

UNIQUE DISTRIBUTED ARCHITECTURE

Distributed Multi-Site Data Flow Architecture



A key aspect of the solution was **Agaram's advanced distributed architecture**, which addressed the issue of unreliable internet connectivity. Thus, each regional lab was provided with its own LIMS instance installed on local servers, enabling them to operate independently of the internet. Key features of this architecture include,

- **Local Autonomy:** Each lab operates independently with minimal reliance on the central system, ensuring that workflows continue uninterrupted even when internet access is unavailable.
- **Central Data Aggregation:** Data from regional labs is synchronized with the central national lab whenever internet connectivity is available. The central lab aggregates data from all regional sites to generate national-level reports on TB cases.

- **Master Data Sharing:** Central labs can share critical master data (e.g., diagnostic standards) with regional labs, ensuring consistency and standardization across the entire network.
- **Real-Time Reporting:** The central server aggregates data from all regional labs to provide real-time TB spread reports, which are essential for national disease surveillance and reporting to the Ministry of Health.

ENABLING SEAMLESS SAMPLE TESTING

Sample testing across geographically distributed laboratories with limited connectivity presented significant challenges in Tajikistan and Uzbekistan's fight against tuberculosis (TB). Laboratories operated with outdated and poorly connected systems, which hindered real-time reporting and efficient disease surveillance.

The deployment of locally hosted Qualis LIMS instances transformed this landscape, ensuring that TB samples could be tested, tracked, and reported smoothly across regions regardless of internet availability. As a result, regional labs were able to maintain uninterrupted testing workflows and ensure timely sample tracking and reporting. When connectivity resumed, seamless data synchronization with the central servers enabled unified, accurate national-level insights. This solution improved sample testing, enhanced data integrity, and significantly strengthened the public health infrastructure for TB control across Central Asia.

CUSTOMIZATION & INTEGRATIONS

Qualis LIMS was customized to meet the specific needs of TB laboratories, ensuring compliance with national health regulations.

The Interfacer Middleware aided in integrating various diagnostic instruments such as GeneXpert, BACTEC MGIT, and BD-Epicenter, and extract precise data.

TRAINING, TESTING, & SUPPORT

Agaram provided extensive training to laboratory staff on using the Qualis LIMS, Interfacer Middleware and Qualis Portal, ensuring they could efficiently operate the systems and handle challenges related to data synchronization or instrument integration.

The system was thoroughly tested to meet the operational and regulatory needs of TB laboratories, including offline operation, data synchronization between regional and central labs, and accurate report generation.

After successful testing, the system was rolled out across all regional and national labs in Tajikistan and Uzbekistan, with Agaram continuing to offer ongoing support for smooth operations, system maintenance, and necessary updates.

KEY BENEFITS

- **Increased Efficiency:** The digitization of TB laboratory workflows has significantly reduced manual processes, leading to faster turnaround times for diagnostic results.
- **Enhanced Data Management:** With Qualis LIMS, laboratories can now manage TB data more effectively, ensuring that all test results and reports are stored in a centralized system for easy access.
- **Real-Time Reporting:** The distributed architecture enables real-time data reporting to national health authorities, allowing for more accurate monitoring of TB cases and faster responses to outbreaks.

- **Resilience to Connectivity Issues:** The system's ability to operate independently of internet access ensures that laboratories can continue their operations without interruptions, even in places with poor connectivity.
- **Seamless Integration:** The integration of diagnostic instruments into the LIMS system has eliminated manual data entry, improved accuracy and reduced errors in laboratory results.
- **Scalability:** The system is designed to scale as more laboratories are brought online, ensuring that new labs can be easily added to the network with minimal disruptions.

This project has played a key role in supporting the national health goals of Tajikistan and Uzbekistan and contributing to the larger mission of the TB Healthcare Service Provider in eliminating TB in Central Asia.

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